AI-BASED INTELLIGENT BUG ANALYSIS AND SECURITY RISK PRIORITIZATION SYSTEM

MINI PROJECT REPORT

Submitted by

BHUVAN SHANKAR P **Register No.: 23UCS027**

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Dr. M. SHANMUGAM

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SRI MANAKULA VINAYAGAR ENGINEERING COLLEGE

(An Autonomous Institution)

MADAGADIPET, PUDUCHERRY - 605107

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SRI MANAKULA VINAYAGAR
ENGINEERING COLLEGE
(AN AUTONOMOUS INSTITUTION)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

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PROJECT GUIDE	PROJECT COORDINATOR	HEAD OF THE DEPARTMENT
[Dr. M. SHANMUGAM]	[Dr. M. SHANMUGAM]	[Dr. N. DANAPAQUIAME]

INTERNAL GUIDE

ABSTRACT

Software development involves handling numerous bug reports that may contain incomplete information, repeated issues, and potential security risks. Analysing these reports manually can be time-consuming and may delay the identification of critical problems. The proposed AI-Based Intelligent Bug Analysis and Security Risk Prioritization System aims to simplify this process by combining data processing, machine learning, deployment readiness, and security analysis within a single application, It helps organise bug reports, identify important issues.

The system collects bug reports through manual entry and CSV file uploads, validates the input, handles missing values, identifies duplicate records, and stores the processed information in a structured database. This provides a consistent foundation for further analysis and makes it easier to manage large collections of software issues. The application is designed to run locally during development, with suitable configuration files, dependency management, and deployment instructions to support future cloud hosting.

Machine learning techniques are used to analyse bug descriptions and predict their severity, category, and affected software module. The TF-IDF technique converts textual information into numerical features, while the Multinomial Naive Bayes algorithm classifies reports using labelled training data. Cosine similarity helps identify related bug reports, reducing repetitive investigation. Model evaluation metrics, including accuracy, precision, recall, and macro F1-score, are used to assess classification performance and understand the reliability of the predictions.

The system also examines bug reports for potential security concerns, including authentication weaknesses, SQL injection indicators, cross-site scripting, and access-control issues. Rule-based screening identifies relevant indicators, while a weighted risk-scoring method helps prioritise reports according to severity, potential impact, security evidence, and uncertainty. The results are presented with explanations to support further investigation and human review. Overall, the proposed system aims to improve bug management, support early identification of potentially serious issues, reduce manual effort, and provide a practical foundation for more efficient and security-aware software maintenance.

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